Identifying and Completing Hebrew Multi-Word Expressions: A Comparative Study of Masked Language Models

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Abstract

The processing and comprehension of Multi-Word Expressions (MWEs), particularly idioms, present a significant challenge in the field of Natural Language Processing (NLP). This challenge arises from the semantic gap between the holistic meaning of the expression and the literal meaning of its constituent words. This project presents a comprehensive comparative study evaluating the ability of modern language models to address this challenge in Hebrew, utilizing a Masked Word Completion task. Three types of models were evaluated: a Hebrew-specific model (AlephBERT), a multilingual model (mBERT), and a Large Language Model (GPT-4o-mini) in Zero-Shot, Few-Shot, and CoT(chain of thoughts) settings. The results demonstrate that AlephBERT exhibits significantly superior performance (achieving 78.7% Top-5 fuzzy accuracy), highlighting the importance of training on a language-specific corpus.

Results reveal distinct differences across model types: AlephBERT excelled in exact completion of idioms, leveraging its exposure to Hebrew morphology and collocations. mBERT performed poorly across all tasks, reflecting insufficient idiomatic representation in multilingual training. GPT showed flexible semantic reasoning, with strong gains from prompt engineering (Few-Shot, MWE-informed, Chain-of-Thought). However, GPT often produced fuzzy matches rather than exact idiom completions, highlighting a trade-off between semantic plausibility and lexical precision.

The findings emphasize the complementary strengths of encoder vs. decoder architectures: encoders capture fixed collocations, while decoders adapt flexibly to semantic variation.

1 Introduction and Problem Definition

Multi-Word Expressions (MWEs), including idioms and fixed phrases, are an integral and rich component of any natural language. These expressions, such as the Hebrew עם יד על הדופק (lit. "with a hand on the pulse," meaning "keeping a close watch"), carry a unique meaning that is not directly derivable from the combination of their constituent words. This semantic gap, known as non-compositionality, poses a significant challenge for Natural Language Processing (NLP) systems, which are typically designed to process language compositionally. The ability to accurately identify, interpret, and complete idioms is crucial for a wide range of applications, from machine translation and sentiment analysis to dialogue systems and reading comprehension. This task serves as a practical simulation for real-world NLP challenges, such as predictive text and autocomplete systems.

1.1 Project Goal

The goal of this project is to systematically evaluate and compare the ability of modern language models, based on the Transformer architecture, to handle the challenge of understanding Hebrew idioms. This evaluation is conducted through a Masked Word Completion task, which requires the model not only to recognize the linguistic pattern of the idiom but also to comprehend the semantic context to select the correct completion.

The project focuses on a comparison between three key approaches:

1. **A Hebrew-Specific Model (AlephBERT):** a Hebrew-specific BERT-base model pre-trained on a large and diverse Hebrew corpus.
2. **A Multilingual Model (mBERT):** a multilingual BERT-base model trained on over 100 languages, including Hebrew.
3. **A Large Language Model (GPT-4o-mini):** a state-of-the-art generative model, evaluated via prompt-based interaction (zero-shot, few-shot, and Chain-of-Thought prompting).

The Hebrew language, being morphologically rich, adds another layer of complexity to this challenge. Inflection of words (e.g., for gender, number, and person) within an idiom can create numerous variations of the same expression, making it difficult to identify through simple pattern matching. For example, the expression דם על הידיים ("blood on the hands") can appear as דמה על ידיה ("her blood on her hands") or דמם על ידיהם ("their blood on their hands").

2 Related Works

2.1 Multiword Expression Identification

MWE identification has long been studied in NLP, traditionally framed as a sequence tagging task (Constant et al., 2017). Early methods relied on linguistic rules; for example, Pasquer et al. (2020) proposed Seen2Seen, which learns verbal MWE candidates and filters them using morphosyntactic constraints.

With the rise of pretrained language models, contextual embeddings became central. Taslimipoor et al. (2020) introduced MTLB-STRUCT, a semi-supervised method fine-tuning BERT to jointly predict VMWEs and syntactic dependencies, evaluated on the PARSEME shared task (Savary et al., 2023).

2.2 Idiomatic Expression Identification

Idioms are a subclass of MWEs characterized by their figurative meaning and lack of compositionality (Baldwin, 2010). Several datasets have been introduced for idiom identification, including VNC-Tokens (Cook et al., 2008), Gigaword (Sporleder and Li, 2009), and SemEval-2013 Task 5 (Korkontzelos et al., 2013).

3 Methodology

This section details the workflow, from data preparation to model evaluation.

3.1 Data Preparation

The project's dataset was constructed from the `train.cupt` file of the PARSEME corpus, which contains annotations for MWEs in Hebrew texts. The file contains 14,035 sentences. The process involved several steps:

1. **MWE Identification and Extraction:** The corpus was scanned to identify sentences annotated with MWEs in the .cupt file (column 11). This process yielded several hundred candidate sentences, from which we selected a subset of multi-word idioms suitable for the masked-word task. After filtering out edge cases (e.g., single-word MWEs), the final dataset comprised 47 idiomatic sentences, each representing a distinct expression, later complemented by literal control examples

2. **Creation of the Idiomatic Dataset:** To test the models' ability to comprehend idioms, we designed a fill-in-the-blank style task. For each idiomatic sentence, the final word of the idiom was masked using the [MASK] token. This masked word was stored as the gold label the model was required to predict.

   For Example:

   - **Original sentence:** יושב ראש הוועדה הודיע שהוועדה תהיה עם יד על הדופק בעניין (The committee chairman announced that the committee will keep a close watch on the matter.)
   - **Masked input:** יושב ראש הוועדה הודיע שהוועדה תהיה עם יד על [MASK] בעניין
   - **Gold label:** הדופק (the pulse)

3. **Creation of the Literal Dataset:** For 18 of the most frequent idioms, new literal control sentences were manually created where the same lexical items appeared in their literal sense. For example, for the idiom על כרעי תרנגולת (lit. on chicken legs, meaning "unstable"), we created:

   - **Literal sentence:** ביקשתי מתכון של אורז על כרעי [MASK] ברוטב עגבניות (I asked for a recipe of rice on [MASK] legs in tomato sauce).

   This dataset allowed us to evaluate whether models rely purely on memorized word associations or demonstrate true semantic sensitivity to context.

This process yielded a unified dataset containing both idiomatic and literal examples, where each sample includes the original sentence, the masked sentence, the correct word for completion, and the type of MWE.

3.2 Models Evaluated

We evaluated encoder-based masked language models (AlephBERT, mBERT) alongside a decoder-based generative model (GPT-4o-mini). While AlephBERT is expected to excel in Hebrew-specific tasks, mBERT provides a multilingual baseline, and GPT introduces prompt-driven reasoning capabilities.

3.3 Experimental Setup

The experiments were framed as a masked word completion task, where each model was tasked with predicting the missing token.

- **For AlephBERT and mBERT:** Evaluated using the standard fill-mask pipeline from the HuggingFace transformers library. For each masked instance, the Top-5 predictions were retrieved.
- **For GPT-4o-mini:** Since this model does not natively support masked-token prediction, we used prompt-based completion strategies. Several prompting styles were tested:
  - **Zero-Shot:** A direct instruction to predict the missing word
  - **Few-Shot (2, 4, 8 examples):** The prompt included multiple example sentences with their correct completions before presenting the target sentence.
  - **Chain-of-Thought (CoT) / MWE-Informed Prompt:** he prompt guided GPT step-by-step, e.g., first recognizing the idiom, explaining its meaning, and then predicting the missing word. This was designed to encourage semantic reasoning rather than surface-level word prediction.

3.4 Evaluation Metrics

To evaluate model performance, we used both exact and fuzzy measures:

- **Top-1 Exact Accuracy:** Percentage of cases where the top-ranked prediction matched the gold label exactly.
- **Top-1 Fuzzy Accuracy:** Percentage of cases where the top prediction was a close lexical match to the gold label, determined using Levenshtein similarity $\geq 85\%$ (to account for morphological variants such as gender or number).
- **Top-5 Fuzzy Accuracy:** Percentage of cases where a fuzzy match appeared in the top 5 predictions.
- **Top-5 Exact Accuracy:** Percentage of cases where the gold label appeared in the top 5 predictions

These metrics allow us to distinguish between strict lexical accuracy and semantic-level appropriateness of the predictions.

4 Results and Analysis

This section presents the results of the experiments and provides an analysis of the performance of the different models according to the defined metrics.

4.1 Overall Performance Comparison

The following table summarizes the performance of the three models on the entire idiomatic dataset. The results for GPT-4o-mini refer to the Few-Shot approach with the MWE-informed and CoT prompt, which yielded the best results for this model.

Table 1: Performance Summary of Models on the Idiomatic Dataset

Method	Top-1 Exact (%)	Top-1 Fuzzy (%)	Top-5 Exact (%)	Top-5 Fuzzy (%)
AlephBERT	61.70	68.09	76.60	**78.72**
GPT-4o-mini	34.04	46.81	59.57	65.96
mBERT	10.64	10.64	12.77	12.77

Several key insights emerge from these results:

- **Superiority of the Language-Specific Model:** AlephBERT significantly outperforms the other two models across all metrics. With an accuracy of 68.09% (Top-1 Fuzzy) and 78.72% (Top-5 Fuzzy), it is evident that its pre-training on a vast Hebrew corpus gives it a distinct advantage in understanding the patterns and nuances of the language.
- **The Challenge for mBERT:** The multilingual model, mBERT, struggles considerably with the task, achieving very low scores. This performance highlights the limitations of a generalist multilingual approach for fine-grained linguistic tasks that require a deep understanding of a specific language.
- **Performance of GPT-4o-mini:** The LLM is positioned between the two BERT models. Despite not being fine-tuned for the task, it achieves reasonable accuracy, especially in the Top-5 metric. This indicates its strong general capabilities, but also shows that a smaller, specialized model can still excel at focused tasks.
- **The Importance of Fuzzy Matching:** For AlephBERT and GPT, switching from exact to fuzzy matching results in a 6%–10% increase in accuracy. For example, AlephBERT predicted **נברא** (masculine form) instead of **נבראה** (feminine form), a minor morphological error that the fuzzy metric corrects. For mBERT, there is no improvement, suggesting its errors are primarily semantic rather than morphological.

4.2 The Impact of Prompt Engineering on GPT-4o-mini

One of the interesting aspects of the experiment was examining the impact of different prompting strategies on GPT-4o-mini's performance. The following table compares the various approaches.

Table 2: Comparison of GPT-4o-mini Performance Under Different Prompting Strategies

Prompting Method	**Top-5 Exact (%)**	**Top-5 Fuzzy (%)**
Zero-Shot	53.19	57.45
Few-Shot (2 examples)	46.81	53.19
Few-Shot (4 examples)	55.32	61.70
Few-Shot (8 examples)	51.06	59.57
Few-Shot (MWE-Informed)	57.45	65.96
Few-Shot (CoT)	**59.57**	**65.96**

The results show that adding examples (Few-Shot) improves performance. Specifically, using 4 examples yielded the best result (61.70% fuzzy accuracy). Interestingly, adding more examples (8) did not lead to further improvement, suggesting a point of saturation (figure 1). The MWE-informed prompt improved performance relative to Zero-Shot but did not reach the level of the 4-shot prompt. This might indicate that the model benefits more from concrete examples than from a meta-linguistic explanation of the task.

4.3 Analysis: Idiomatic vs. Literal Context Performance

One of the most central and surprising findings of this study is the comparison of model performance on idiomatic versus literal sentences.

Table 3: Comparison of Accuracy (Fuzzy, Top-1) on Idiomatic vs. Literal Sentences

Model	**Idiomatic Accuracy (%)**	**Literal Accuracy (%)**
AlephBERT	68.09	66.67
GPT-4o-mini	38.30	55.56
mBERT	10.64	5.56

The results reveal a fascinating phenomenon: the two BERT models, particularly AlephBERT, perform **better** on idiomatic sentences than on literal ones. This finding is counter-intuitive, as one might expect models to struggle more with figurative language. The most plausible explanation is that the models heavily rely on strong statistical patterns (collocations) learned from the corpus. An idiom is a fixed pattern; when AlephBERT sees the sequence corresponding to עם יד על, the statistical probability of the next word being הדופק is immense.

In contrast, in a literal sentence meaning "he placed his hand on the [MASK]," the context is much more open, and the completion could be שולחן (table), ספר (book), or כתפו (his shoulder). This task requires a deeper and more flexible semantic understanding of the situation and is more challenging than completing a fixed pattern. Interestingly, GPT-4o-mini, while also showing some performance drop, copes better with the literal contexts, suggesting more flexible semantic reasoning abilities compared to the BERT models.

4.4 Error Analysis

An analysis of the models' errors provides further insights.

- **AlephBERT:** Most errors are "near misses" slight morphological mistakes (gender, number) or substitutions with a semantic synonym (e.g., עומס for נטל). Some are corrected by the fuzzy metric.
- **mBERT:** Errors are typically semantic and out of context. The model tends to complete with common function words or punctuation, indicating a shallow understanding of the sentence.
- **GPT-4o-mini:** GPT's errors are the most interesting. It often provides a completion that is semantically plausible in the broader context but ignores the fixed pattern of the idiom. For

instance, in a sentence meaning "...until [MASK] smoke comes out of the talks...", instead of לבן (white), it suggested מבעד (through), a logical completion that misses the idiomatic pattern. This reinforces the idea that the model prioritizes general semantic reasoning over recalling specific collocations.

Error distribution by MWE type (figure 3):

- Dataset composition: VID = 80.9%, LVC.full = 12.8%, others = 6.3%.
- GPT errors: VID = 87%, LVC.full = 12.5%.

No systematic bias was detected. Instead, the error profile largely mirrors the data distribution, though GPT tended to struggle most with Verb–Idiom Constructions (VIDs), consistent with their dominance in the dataset.

5 Discussion and Conclusions

This project aimed to comparatively evaluate the ability of modern language models to handle Hebrew idioms. The findings provide a multi-faceted view of the state of the field and offer several important insights.

5.1 Summary of Key Findings

The clearest finding is the superiority of the AlephBERT model. Its performance, significantly higher than that of mBERT and GPT-4o-mini, underscores the critical importance of pre-training on a dedicated, extensive corpus for the target language. While multilingual models offer a general solution, they do not achieve the precision required for nuanced, language-specific tasks.

Second, the study demonstrated the contribution of Prompt Engineering for large language models. The Few-Shot approach, particularly with 4 examples, markedly improved GPT-4o-mini performance, showing that it is possible to "teach" the model the task without costly fine-tuning.

The third finding from this research is the performance difference between idiomatic and literal contexts. Contrary to intuition, the BERT models excelled at completing idioms (fixed patterns) more than literal sentences (open contexts). This phenomenon reveals that the models' success stems largely from their ability to learn and recall frequent statistical sequences, rather than from a deep and flexible semantic understanding. GPT-4o-mini, in contrast, showed slightly more flexibility in literal contexts, hinting at more advanced reasoning capabilities.

5.2 Novelty and Contribution

The experiments were conducted using a fill-mask task: for each masked sentence, the models were required to predict the missing word. The primary contribution of this work lies in its novel diagnostic methodology and the insightful analysis it offers into the strengths and limitations of state-of-the-art language models when faced with a nuanced linguistic challenge. The key innovations introduced in this study are:

1. **Empirical Evidence of the "Curse of Context" in MWE Processing.** A key and counter-intuitive finding of this study is that models, particularly the Hebrew-specific AlephBERT, paradoxically achieve higher accuracy on predictable idiomatic phrases than on syntactically similar but semantically open-ended literal sentences. We term this the "curse of context." This result provides strong empirical evidence that models' success in this domain is driven more by their mastery of learned statistical collocations than by flexible semantic disambiguation. This insight into the models' limitations is crucial for understanding their failure modes and guiding future development.

2. **A Systematic Cross-Architectural Benchmark for Hebrew MWEs.** This work presents one of the first direct, systematic comparisons of three distinct model architectures for the task of Hebrew MWE completion: a language-specific encoder (AlephBERT), a multilingual encoder (mBERT), and a frontier Large Language Model (GPT-4o-mini). By evaluating these models on the same controlled dataset, and further probing the LLM with various prompting strategies (Zero-Shot, Few-Shot, and Chain-of-Thought prompts), we provide a clear performance benchmark. The results highlight the current trade-offs between specialized, corpus-specific knowledge and the emergent, general reasoning abilities of LLMs when applied to a morphologically complex language.

5.3 Limitations and Future Work

This study has several limitations to acknowledge. First, the manually created literal dataset was relatively small. Expanding this dataset would allow for more statistically robust conclusions. Second, the task focused solely on completing the final word of the idiom. It would be interesting to examine model performance when other parts of the idiom are masked.

Future research directions could include:

- **Additional models:** should also include evaluating more decoder-only models on the idiom completion task, such as Gemini.
- **Data augmentation:** using synthetic idioms generated by LLMs.
- **Prompt diversity:** including Self-Consistency (SC) and alternative reasoning prompts.
- **More Data:** Larger-scale idiom evaluation beyond the limited PARSEME subset

In conclusion, this project provides a comprehensive and in-depth analysis of the challenge of understanding Hebrew idioms by modern language models. The findings highlight the advantage of language-specific models, the potential of prompt engineering, and reveal the important distinction between pattern recognition and semantic reasoning.

6 Code

The complete code for this project, including data analysis and experiment execution, is available in the Colab Notebook at the following link:
Link to Colab Notebook

References

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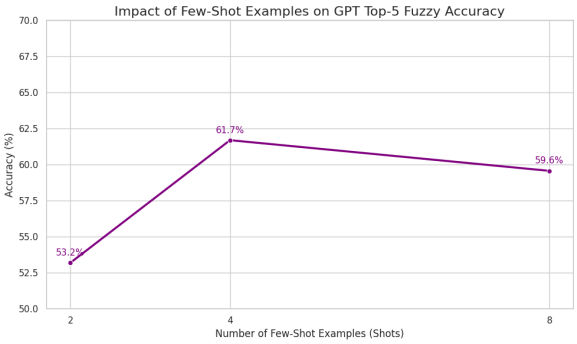
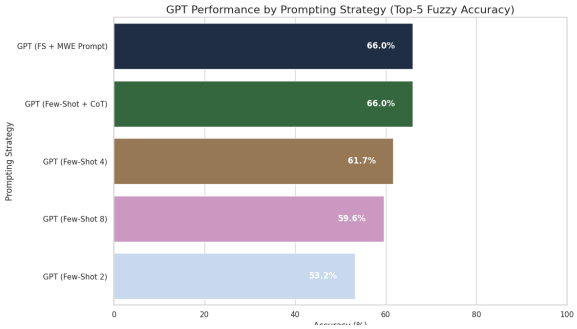
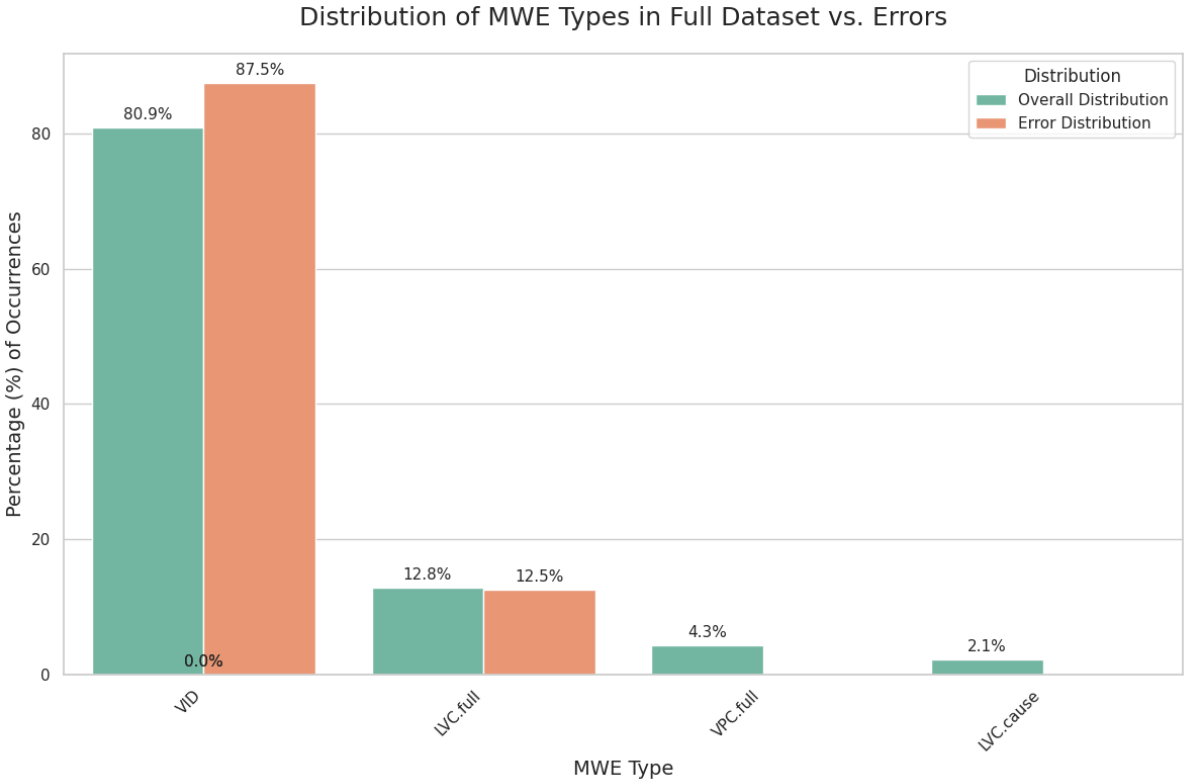
(a) figure 1: GPT Top-5 Fuzzy Accuracy with 2, 4, and 8 few-shot examples



(b) figure 2: GPT performance across prompting strategies (Top-5 Fuzzy Accuracy)



(c) figure 3: Comparison of MWE type distribution in the dataset and in GPT errors